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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Assessment Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 19th May 2016

General Instructions

- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 73 boys

Examiner

DWH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The parametric equations of an ellipse are:

- (A) $x = a \cos \theta$ and $y = b \tan \theta$
- (B) $x = a \cos \theta$ and $y = b \sin \theta$
- (C) $x = a \sec \theta$ and $y = b \tan \theta$
- (D) $x = a \sec \theta$ and $y = b \sin \theta$

QUESTION TWO

Determine $\int \frac{1}{\sqrt{16 - (x + 3)^2}} dx$.

- (A) $\frac{1}{4} \sin^{-1} \left(\frac{x + 3}{4} \right) + C$
- (B) $\frac{1}{4} \tan^{-1} \left(\frac{x + 3}{4} \right) + C$
- (C) $\sin^{-1} \left(\frac{x + 3}{4} \right) + C$
- (D) $\tan^{-1} \left(\frac{x + 3}{4} \right) + C$

QUESTION THREE

The number $2(\cos \pi - i \sin \pi)$ is:

- (A) rational
- (B) undefined
- (C) irrational
- (D) purely imaginary

QUESTION FOUR

A polynomial $P(x)$ with real coefficients has odd degree. Which of the following sentences must be correct?

- (A) The minimum number of real zeroes of $P(x)$ is 1.
- (B) The minimum number of real zeroes of $P(x)$ is 0.
- (C) The minimum number of non-real zeroes of $P(x)$ is 1.
- (D) The minimum number of real zeroes of the derivative $P'(x)$ is 1.

QUESTION FIVE

The equation $x^4 + 2x^3 + 8x + 16 = 0$ has a double root at:

- (A) $x = 1 - \sqrt{3}i$
- (B) $x = 1 + \sqrt{3}i$
- (C) $x = -2$
- (D) $x = 2$

QUESTION SIX

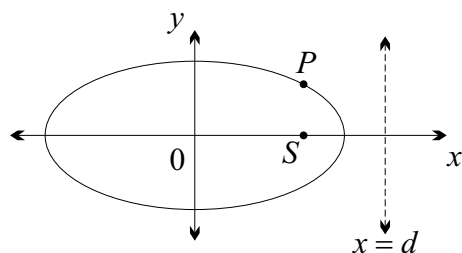
Consider the integral $I = \int_{-2}^4 x^3 \sqrt{16 - x^2} \, dx$. Which is a true statement?

- (A) $I = \int_2^4 x^3 \sqrt{16 - x^2} \, dx$
- (B) $I = 2 \int_0^2 x^3 \sqrt{16 - x^2} \, dx + \int_2^4 x^3 \sqrt{16 - x^2} \, dx$
- (C) $I = \int_{-4}^{-2} x^3 \sqrt{16 - x^2} \, dx$
- (D) $I = \int_{-4}^{-2} x^3 \sqrt{16 - x^2} \, dx + 2 \int_{-2}^0 x^3 \sqrt{16 - x^2} \, dx$

QUESTION SEVEN

A polynomial $P(x)$ has real coefficients and $P(3i) = 0$. Which of the following must be true?

- (A) $P(x)$ has a quadratic factor that has no real roots.
- (B) $P(3) = i$.
- (C) $P((3i)^2) = 0$.
- (D) $P(x)$ is a polynomial of odd degree.

QUESTION EIGHT

An ellipse centred at the origin has a focus at $S(3, 0)$ and a directrix at $x = d$, where $d > 0$. The ellipse also passes through the point $P\left(3, \frac{16}{5}\right)$.

Find d if the eccentricity of the ellipse is $\frac{3}{5}$.

- (A) $d = \frac{123}{25}$
- (B) $d = \frac{48}{25}$
- (C) $d = \frac{25}{3}$
- (D) $d = \frac{13}{3}$

QUESTION NINE

The following graph displays the distance x of a particle from a fixed point O over time t .

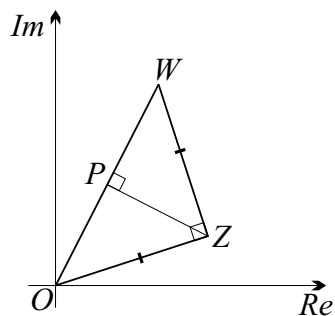


Which of the following could NOT describe a possible motion of the particle for $t \geq 0$?

- (A) The particle is at rest.
- (B) The acceleration of the particle is constant.
- (C) The particle is undergoing uniform circular motion about O .
- (D) The path of the particle is a parabola.

QUESTION TEN

The point Z represents the complex number z . The intervals OZ and ZW are equal in length and perpendicular, as shown. The point P is the foot of the altitude from Z to OW .



The point P represents which complex number?

- (A) $\frac{1}{2}iz$
- (B) $\frac{1}{2}iz^2$
- (C) $\frac{1}{2}(z + iz)$
- (D) $\frac{1}{2}(z - iz)$

————— End of Section I —————

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a) Find $\int \frac{x^2}{\sqrt{1-x^3}} dx$. 2

(b) Express the polynomial $P(x) = x^3 - 5x^2 + 11x - 15$ as a product of linear factors. 3

(c) (i) Expand $(x - 1 + i)(x - 1 - i)$. 1

(ii) Consider the polynomial $P(x) = x^3 + kx - 5$, where k is real. The remainder when $P(x)$ is divided by $x^2 - 2x + 2$ is $9x - 9$. Find k . 2

(d) (i) Find values of A , B and C , such that 2

$$\frac{4x^2 + 11}{(2x + 3)(x + 4)} = A + \frac{B}{2x + 3} + \frac{C}{x + 4}.$$

(ii) Hence, or otherwise, find $\int_{-1}^1 \frac{4x^2 + 11}{(2x + 3)(x + 4)} dx$. 2

(e) A block of mass 4 kg moves in a straight line across a flat surface. At time t seconds its displacement from a fixed origin is x metres and its velocity is v metres per second. The variable force acting on the block is $18 - 8x$ Newtons. When $x = 4$, $v = 2$. 3

Find v^2 in terms of x .

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**(a) Consider the hyperbola $x = 3 \sec \theta$, $y = 2 \tan \theta$.

(i) Find the Cartesian equation of the hyperbola.

1

(ii) Find the equations of the asymptotes of the hyperbola.

1

(iii) Find the foci of the hyperbola.

2

(iv) Sketch the hyperbola, showing asymptotes and any intercepts with axes.

2

(b) An object of mass m is dropped and is then subject to gravity and air resistance. When its displacement is x metres, its velocity is $v \text{ ms}^{-1}$. The magnitude of forces acting on the object are gravity mg Newtons, and air resistance kv^2 Newtons, for some positive constant k . Take downwards as positive.

2

Write down an expression for the acceleration of the object and hence find an expression for the terminal velocity V_T .

(c) Use the substitution $x = 2 \sin \theta$ to find $\int_0^1 \frac{1}{(4-x^2)^{\frac{3}{2}}} dx$.

3

(d) Consider the polynomial $P(x) = x^3 - 5x^2 - 2x - 8$ with zeroes α , β and γ .

(i) Find a simplified polynomial expression $Q(x)$ with zeroes $\frac{\alpha}{2}$, $\frac{\beta}{2}$ and $\frac{\gamma}{2}$.

2

(ii) Hence, or otherwise, find a polynomial $T(x)$ with zeroes

2

$$\frac{3\alpha}{2} + \beta + \gamma, \alpha + \frac{3\beta}{2} + \gamma \text{ and } \alpha + \beta + \frac{3\gamma}{2}.$$

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) You may assume the equation of the tangent to a hyperbola is $\frac{x_1x}{a^2} - \frac{y_1y}{b^2} = 1$ and the equation of the chord of contact is $\frac{x_0x}{a^2} - \frac{y_0y}{b^2} = 1$. Do NOT prove these equations.

For the hyperbola $\frac{x^2}{225} - \frac{y^2}{144} = 1$, find:

- (i) the chord of contact from the point $(15, 6)$, and **1**
 (ii) the two tangents to the hyperbola passing through the point $(15, 6)$. **3**

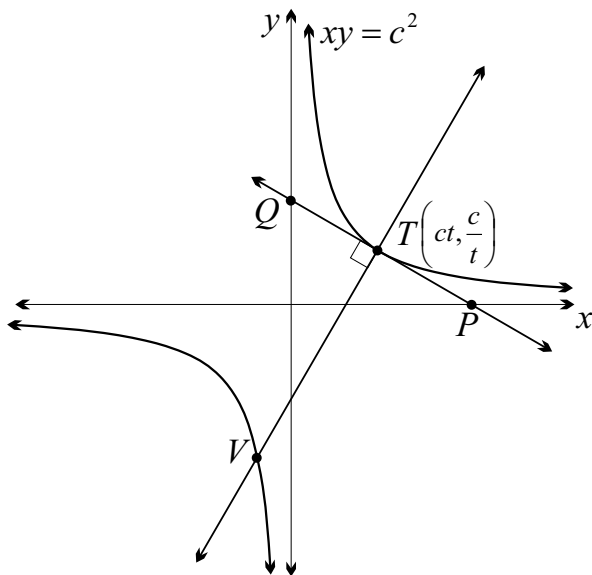
- (b) Consider the integral $I_n = \int_0^1 (1+x^2)^n dx$, where n is a non-negative integer. Use **3**
 integration by parts to show that for $n \geq 1$, $I_n = \frac{2^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$.

- (c) Consider the polynomial equation $x^3 - 3x^2 - 2x - 1 = 0$ with roots α , β and γ .

(i) Find the polynomial equation with roots α^2 , β^2 and γ^2 . **2**

(ii) Find $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$. **1**

- (d) The tangent to the hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ meets the x -axis at P and the y -axis at Q . The normal at T meets the other arm of the hyperbola at V .



You may use the equation for the tangent at T given by $x + t^2y = 2ct$ and the equation for the normal at T given by $t^2x - y = ct^3 - \frac{c}{t}$. Do NOT prove these.

- (i) Find the co-ordinates of P and Q . **1**
 (ii) Show that the co-ordinates of V are $\left(-\frac{c}{t^3}, -ct^3\right)$. **2**
 (iii) Show that $\triangle PQV$ is isosceles. **2**

Examination continues next page ...

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Show that the equation of the normal to an ellipse at the point
- $(a \cos \theta, b \sin \theta)$
- is given
- 3**

$$\text{by } \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2.$$

- (b) A ball of unit mass is projected vertically upwards from ground level with initial velocity
- $U \text{ ms}^{-1}$
- .

Take the point of launch as the origin. At t seconds after launch, the displacement from the origin is x metres and the velocity is $v \text{ ms}^{-1}$. Take upward motion as positive.

The resistive force due to its passage through the air is proportional to the velocity of the ball and the equation of motion is $\ddot{x} = -kv - g$ for some constant $k > 0$.

- (i) Show that the greatest height achieved is
- $H = \frac{U}{k} - \frac{g}{k^2} \log_e \left(\frac{g + kU}{g} \right)$
- .
- 2**

The ball reaches its maximum height and begins to fall back towards the ground.

Assume that at time t seconds after the ball starts to fall, the displacement is y metres and the velocity is $w \text{ ms}^{-1}$, from the point at which it begins to fall. Take downward motion as positive. The equation of motion is $\ddot{y} = -kw + g$.

- (ii) It hits the ground when
- $y = H$
- . Show that in terms of the velocity
- $W \text{ ms}^{-1}$
- at which the ball hits the ground,
- 2**

$$H = \frac{-W}{k} + \frac{g}{k^2} \log_e \left(\frac{g}{g - kW} \right).$$

- (iii) Let
- $T = \frac{g}{k}$
- be the terminal velocity and
- $U_T = \frac{U}{T}$
- and
- $W_T = \frac{W}{T}$
- be the ratios of the launch and impact speeds to the terminal velocity respectively. Show that
- 2**

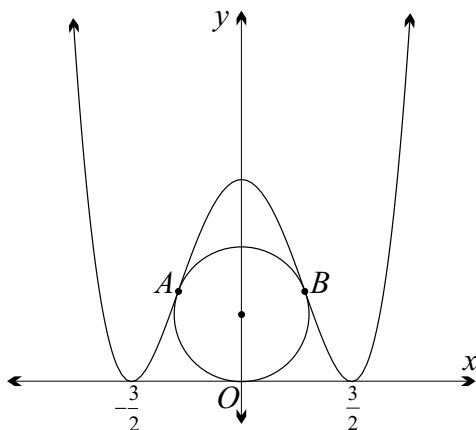
$$U_T + W_T = \log_e \left(\frac{1 + U_T}{1 - W_T} \right).$$

- (iv) Show by substitution that if the ball is thrown at 50% of the terminal velocity then it will impact at approximately 37% of the terminal velocity.
- 1**

Exam continues on the next page

QUESTION FOURTEEN (Continued)

- (c) The circle $x^2 + (y - r)^2 = r^2$ is tangent to the curve $y = \left(x^2 - \frac{3}{2}\right)^2$ at two points A and B , as shown.



- (i) Show that at the points of contact, $u^4 - 2u^2r + u + \frac{3}{2} = 0$, where $u = x^2 - \frac{3}{2}$. 1
- (ii) Explain why $4u^3 - 4ur + 1 = 0$. 1
- (iii) There is a single real solution for u , for which $|x| < \frac{3}{2}$. Find this solution. 2
- (iv) Find the radius of the circle. 1

_____ End of Section II _____

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

SOLUTIONS

(1) (B)

(2) (C)

(3) $2(\cos \pi - i \sin \pi)$

$= 2(1 - 0i)$

$= 2$

$\therefore$ (A) rational

(4)

For a polynomial with odd degree, if the leading coefficient, a is positive $a > 0$, as $x \rightarrow -\infty$, $y \rightarrow -\infty$ and as $x \rightarrow \infty$, $y \rightarrow \infty$ and therefore (given it is continuous) it will cut the x -axis for some $x \in \mathbb{R}$

or $a < 0$, as $x \rightarrow -\infty$, $y \rightarrow \infty$ and as $x \rightarrow \infty$, $y \rightarrow -\infty$ and similarly...

$\therefore$ (A) is correct.

(B) wrong (since A is correct)

(C) Certainly it's possible to have zero non-real zeros. e.g. $P(x) = (x-2)^3$

(D) ~~For~~ For example: $P'(x) = x^2 + 3$ has no real roots

$$P(x) = \frac{x^3}{3} + 3x + 2$$

$\therefore$ D is wrong.

(5)

$$\text{Let } P(x) = x^4 + 2x^3 + 8x + 16$$

$$\begin{aligned}\therefore P'(x) &= 4x^3 + 6x^2 + 8 \\ &= 2(2x^3 + 3x^2 + 4)\end{aligned}$$

$$P'(2) > 0$$

$$P'(-2) = 2(-16 + 12 + 4) = 0$$

$$\text{now } P(-2) = 16 - 16 - 16 + 16 = 0$$

$\therefore -2$ is a double root. $\therefore$ (C)

$$A: P(1 - \sqrt{3}i) = 0, \text{ but } P'(1 - \sqrt{3}i) = -36 - 12\sqrt{3}i \neq 0$$

$$B: P(1 + \sqrt{3}i) = 0, \text{ but } P'(1 + \sqrt{3}i) = -36 + 12\sqrt{3}i \neq 0$$

$$D: \underline{P(2) > 0 \neq 0}.$$

(6)

$$I = \int_{-2}^4 x^3 \sqrt{16 - x^2} dx$$

$$\begin{aligned}\text{if } f(x) &= x^3 \sqrt{16 - x^2} \\ f(-x) &= -x^3 \sqrt{16 - x^2} \\ &= -f(x)\end{aligned}$$

$$= \int_{-2}^2 x^3 \sqrt{16 - x^2} dx + \int_2^4 x^3 \sqrt{16 - x^2} dx$$

$$= 0 + \int_2^4 x^3 \sqrt{16 - x^2} dx$$

$\therefore$ (A)

(B) would be true if f were an even fn.

(7)

since $P(x)$ has real coefficients, its complex zeros must come in conjugate pairs.

$$P(3i) = 0 \quad \therefore P(-3i) = 0$$

$$\therefore (x - 3i)(x + 3i) \text{ is a factor}$$

$$\therefore (x^2 + 9) \text{ is a factor, which is quadratic with no real roots}$$

$$\therefore \textcircled{A}$$

(8)

Easiest method:

$$\frac{PS}{PQ} = e$$

(Q is closest point on directrix to P)

$$\frac{\frac{16}{5}}{\frac{5}{3}} = \frac{3}{5}$$

$$\therefore PQ = \frac{16}{5} \times \frac{5}{3} = \frac{16}{3}$$

$$\therefore Q \text{ will be } \left(3 + \frac{16}{3}, \frac{16}{5}\right)$$

$$\therefore \text{directrix is } x = \underline{\underline{\frac{25}{3}}}$$

$\textcircled{C}$

(9)

$\textcircled{A}$

Certainly possible

$\textcircled{B}$

acceleration = 0 is possible

$\textcircled{C}$

certainly possible, since a circle of radius 4 makes constant displacement

$\textcircled{D}$ Not possible

(10)

$$\overrightarrow{OZ} = z, \quad \overrightarrow{ZW} = iz$$

$$\overrightarrow{OW} = z + iz$$

$$\overrightarrow{OP} = \frac{1}{2} \overrightarrow{OW} \quad (\text{by symmetry of triangle } OZW)$$

$$\therefore P = \frac{1}{2}(z + iz)$$

$\textcircled{D}$

$\textcircled{C}$

11 (a) $\int \frac{x^2}{\sqrt{1-x^3}} dx = -\frac{1}{3} \int (-3x^2) (1-x^3)^{-\frac{1}{2}} dx$ (1)

or substⁿ
 $u = 1-x^3$
 or similar

$$= -\frac{1}{3} (1-x^3)^{-\frac{1}{2}} \times 2 + C$$
 (2)

$$= -\frac{2}{3} \sqrt{1-x^3} + C$$

Progress: (1)

Answer: (2)

(b) $P(x) = x^3 - 5x^2 + 11x - 15$

$$P(1) = 1 - 5 + 11 - 15 \neq 0$$

$$P(-1) = -1 - 5 - 11 - 15 \neq 0$$

$$P(3) = 27 - 45 + 33 - 15 = 0$$
 (1)

$\therefore x-3$ is a factor

$$\begin{array}{r} x^2 - 2x + 5 \\ x-3 \overline{) x^3 - 5x^2 + 11x - 15} \\ \underline{x^3 - 3x^2} \\ -2x^2 \\ \underline{-2x^2 + 6x} \\ 5x \\ \underline{5x - 15} \\ 0 \end{array}$$

$$P(x) = (x-3)(x^2-2x+5)$$
 (2)

Now, $x^2 - 2x + 5 = (x-1)^2 + 2^2$

$$= (x-1+2i)(x-1-2i)$$

$$\therefore P(x) = (x-3)(x-1+2i)(x-1-2i)$$
 (3)

(c) (i) ~~$(x-(1-i))(x-(1+i)) = x^2 - 2x + 2$~~

$$(x-1+i)(x-1-i) = (x-1)^2 - i^2 = x^2 - 2x + 1 + 1$$

$$= x^2 - 2x + 2$$
 (1)

(ii) given (i) $1+i$ and $1-i$ are ~~zeros~~ zeros of $x^2 - 2x + 2$

also $P(x) = x^3 + kx - 5 = (x^2 - 2x + 2)Q(x) + 9x - 9$

$$(1) (c) (ii) P(x) = x^3 + kx - 5 = (x^2 - 2x + 2) Q(x) + 9x - 9 \quad (1)$$

$$P(1+i) = (1+3i-3-i) + k(1+i) - 5 = 0 \times Q(1+i) + 9 + 9i - 9$$

$$\text{Since } k \in \mathbb{R} \quad -7+k=0 \quad \text{And } i(2+k) = 9i \\ \therefore k=7 \quad \therefore k=7 \quad (2)$$

$$\underline{\text{OR}} \quad P(1-i) = (1-3i-3+i) + k(1-i) - 5 = 0 \times Q(x) + 9 - 9i - 9 \\ -7+k + (2-k)i = -9i \\ \underline{k=7} \quad \text{and} \quad \underline{k=7}$$

By Inspection, $A=2$

$$(d) (i) \frac{x^2+11}{(2x+3)(x+4)} = \frac{2(2x^2+11x+12)}{2x^2+11x+12} + \frac{-22x-13}{(2x+3)(x+4)}$$

$$\therefore -22x-13 = B(x+4) + C(2x+3) \quad (1)$$

$$B+2C = -22 \quad \dots (1)$$

$$4B+3C = -13 \quad \dots (2)$$

$$4 \times (1) - (2) \quad 0B+5C = -88+13 = -75$$

$$\underline{C = -15}$$

$$\text{sub in (1)} \quad B-30 = -22 \\ \underline{B=8} \quad (2)$$

$$(ii) \int_{-1}^1 \frac{x^2+11}{(2x+3)(x+4)} dx = \int_{-1}^1 \left(2 + \frac{8}{2x+3} - \frac{15}{x+4} \right) dx$$

$$= \left[2x \right]_{-1}^1 + 8 \left[\ln(2x+3) \right]_{-1}^1 - 15 \left[\ln(x+4) \right]_{-1}^1 \quad (1)$$

$$= 4 + 8 \ln 5 - 8 \ln 1 - 15 \ln 5 + 15 \ln 3$$

$$= 4 + 15 \ln 3 - 7 \ln 5$$

$$\underline{\hspace{10em}} \quad (2)$$

11(e) -

$$F = 18 - 8x = m\ddot{x} = 4\ddot{x}$$

$$\ddot{x} = \frac{9}{2} - 2x$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{9}{2} - 2x \quad (1)$$

$$\frac{1}{2} v^2 = \frac{9}{2} x - x^2 + C_1 \quad \text{~~WR~~}$$

$$v^2 = 9x - 2x^2 + C_2 \quad (2)$$

$$x=4, v=2: \quad 4 = 36 - 32 + C_2 \Rightarrow \underline{C_2 = 0}$$

must be evaluated
for (2) marks to
be awarded.

$$\therefore \underline{v^2 = 9x - 2x^2} \quad (3)$$

12

(a) $x = 3 \sec \theta$, $y = 2 \tan \theta$

(i) $\sec^2 \theta = 1 + \tan^2 \theta$

$$\left(\frac{x}{3}\right)^2 = 1 + \left(\frac{y}{2}\right)^2$$

$$\frac{x^2}{9} - \frac{y^2}{4} = 1 \quad (1)$$

(ii) asymptotes at $y = \pm \frac{2}{3}x$ (1)

(iii) $b^2 = a^2(e^2 - 1)$ Focus $(ae, 0)$ or $(-ae, 0)$

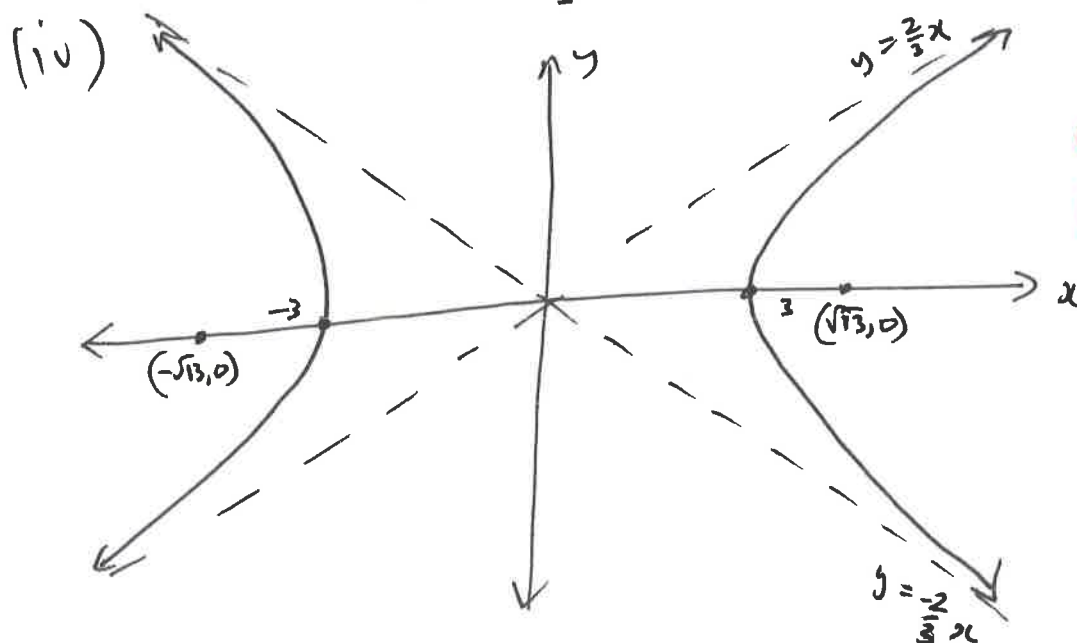
$$\frac{4}{9} = e^2 - 1$$

$$e^2 = \frac{13}{9}$$

$$e = \frac{\sqrt{13}}{3} \quad (1)$$

$$= \left(\frac{\sqrt{13}}{3} \times 3, 0\right) \text{ or } \left(-\frac{\sqrt{13}}{3} \times 3, 0\right)$$

$$= (\sqrt{13}, 0) \text{ or } (-\sqrt{13}, 0) \quad (2)$$



(2) marks
Feature omitted
or incorrect
according to
student's
working (1).

(b) $\ddot{x} = -g + kv^2$

$$0 = -g + k V_T^2$$

$$V_T^2 = \frac{g}{k}$$

$$V_T = \sqrt{\frac{g}{k}} \quad (2)$$

at $\ddot{x} = 0$, $v = V_T$ (1)

This solution was to a previous draft of the question. The corrected solution follows.

Corrected solution for Q12 (b):

(b)

$$m \ddot{x} = +mg - kv^2$$

$$\ddot{x} = g - \frac{k}{m}v^2 \quad \textcircled{1} \text{ terminal velocity when } \underline{\ddot{x}=0}$$

$$0 = g - \frac{k}{m}V_T^2$$

$$V_T^2 = \frac{gm}{k}$$

$$V_T = \sqrt{\frac{gm}{k}} \quad \textcircled{2} \text{ (since down is positive, } V_T > 0)$$

12(c)

$$x = 2 \sin \theta$$

$$dx = 2 \cos \theta d\theta$$

$$x=0, \theta=0$$

$$x=1, \sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\int_0^1 \frac{1}{(4-x^2)^{3/2}} dx = \int_0^{\pi/6} \frac{2 \cos \theta d\theta}{(4(1-\sin^2 \theta))^{3/2}} \quad (1)$$

$$= \frac{1}{4} \int_0^{\pi/6} \frac{\cos \theta}{\cos^3 \theta} d\theta$$

$$= \frac{1}{4} \int_0^{\pi/6} \sec^2 \theta d\theta$$

$$= \frac{1}{4} \left(\tan \theta \right)_0^{\pi/6} \quad (2)$$

$$= \frac{1}{4} \times \frac{1}{\sqrt{3}} = \frac{1}{4\sqrt{3}} \quad (3)$$

$$12(d) \quad P(x) = \cancel{5x} x^3 - 5x^2 - 2x - 8$$

(i) $P(2x)$ will have zeros $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$ (1)

$$P(2x) = 8x^3 - 20x^2 - 4x - 8$$

$$Q(x) = 2x^3 - 5x^2 - x - 2 \quad (\text{simplified}) \quad (2)$$

(ii) $\frac{3\alpha}{2} + \beta + \gamma, \alpha + \frac{3\beta}{2} + \gamma, \alpha + \beta + \frac{3\gamma}{2}$ are equal to:

$$(\alpha + \beta + \gamma) + \frac{\alpha}{2}, (\alpha + \beta + \gamma) + \frac{\beta}{2}, (\alpha + \beta + \gamma) + \frac{\gamma}{2} \quad (1)$$

$$\text{now } \alpha + \beta + \gamma = -(-5) = 5 \quad (\text{from } P(x))$$

$$\therefore \text{Required zeros are } \frac{\alpha}{2} + 5, \frac{\beta}{2} + 5, \frac{\gamma}{2} + 5$$

$\therefore Q(x-5)$ will give these zeros (2)

$$\begin{aligned} Q(x-5) &= 2(x-5)^3 - 5(x-5)^2 - (x-5) - 2 \\ &= 2x^3 - 30x^2 + 150x - 250 \\ &\quad - 5x^2 + 50x - 125 \\ &\quad - x + 5 \\ &\quad - 2 \end{aligned}$$

$$\therefore T(x) = 2x^3 - 35x^2 + 149x - 372 \quad (3)$$

$$(13a) \frac{x^2}{225} - \frac{y^2}{144} = 1 \quad a=15 \quad b=12$$

$$216x - 135y = 3240$$

(i) Chord of contact @ (15, 6)

$$\frac{15x}{225} - \frac{6y}{144} = 1$$

$$\frac{x}{15} - \frac{y}{24} = 1$$

$$\text{OR } \frac{y = \frac{8}{5}(x-15)}{y = \frac{8x}{5} - 24}$$

$$\text{OR } 8x - 5y - 120 = 0$$

(ii) Solve Chord of Contact & Hyp Simultaneously $x = \frac{5}{8}(y+24)$

$$y = \frac{8}{5}(x-15) = \frac{8}{5}(x-24)$$

$$x = \frac{5}{8}y + 15 = \frac{120+5y}{8}$$

$$\therefore \frac{x^2}{15^2} - \frac{8^2(x-15)^2}{5^2 \times 12^2} = 1 \quad \checkmark$$

$$\frac{x^2}{15^2} = \left(1 + \frac{y}{24}\right)^2$$

$$(\times 15^2) \quad x^2 - 4(x-15)^2 = 225$$

$$x^2 - 4(x^2 - 30x + 225) = 225$$

$$-3x^2 + 120x - 900 = 225$$

$$3x^2 - 120x + 5 \times 225 = 0$$

$$x^2 - 40x + 5^2 \times 15 = 0$$

$$(x-25)(x-15) = 0 \quad \checkmark$$

$$x=15 \quad \text{OR} \quad x=25$$

$$y=0 \quad y=16$$

$$\text{ie } \frac{15x}{225} = 1$$

$$x=15$$

Vertical

$$\frac{25x}{225} - \frac{16y}{144} = 1$$

$$\frac{x}{9} - \frac{y}{9} = 1$$

$$x - y - 9 = 0 \quad \checkmark$$

$$\text{OR } y = x - 9$$

$$\left(1 + \frac{y}{24}\right)^2 - \frac{y^2}{12^2} = 1$$

$$\frac{1}{24^2}(24+y)^2 - \frac{y^2}{24^2} = 1$$

$$24^2 + 48y + y^2 - 4y^2 = 24^2$$

$$48y - 3y^2 = 0$$

$$3y(16-y) = 0$$

$$(13)(b) \quad I_n = \int_0^1 (1+x^2)^n dx$$

$$dv = dx \\ v = x$$

$$u = (1+x^2)^n$$

$$du = n(1+x^2)^{n-1} \cdot 2x dx$$

$$\therefore I_n = \left[(1+x^2)^n x \right]_0^1 - \int_0^1 x \cdot n(1+x^2)^{n-1} \cdot 2x dx \quad (1)$$

$$= (2^n - 0) - 2n \int_0^1 x^2 (1+x^2)^{n-1} dx$$

NB ~~2~~

$$\boxed{x^2 = (1+x^2) - (1)}$$

$$= 2^n - 2n \left[\int_0^1 (1+x^2)(1+x^2)^{n-1} dx - \int_0^1 (1+x^2)^{n-1} dx \right] \quad (2)$$

$$= 2^n - 2n (I_n - I_{n-1})$$

$$I_n (1+2n) = 2^n + 2n I_{n-1} \quad (3)$$

$$I_n = \frac{2^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$$

c) $P(x) = x^3 - 3x^2 - 2x - 1 = 0$ roots α, β, γ

(i) Replace x with $\sqrt{x}$

$$P(\sqrt{x}) = x\sqrt{x} - 3x - 2\sqrt{x} - 1 = 0 \quad \checkmark$$

$$\therefore \sqrt{x}(x-2) = 3x+1$$

$$x(x^2 - 4x + 4) = 9x^2 + 6x + 1 \quad \checkmark$$

$$\underline{x^3 - 13x^2 - 2x - 1 = 0}$$

(ii) $\Sigma \alpha\beta = q_a = (-2)$ $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \underline{(-2)} \quad \checkmark$

d) (i) Tangent $x + t^2 y = 2ct$
 at P $y=0 \therefore x=2ct$
 at Q $x=0 \therefore t^2 y = 2ct$
 $y = \frac{2c}{t}$

$$\frac{P(2ct, 0)}{Q(0, \frac{2c}{t})} \quad \checkmark$$

(ii) Normal $t^2 x - y = ct^3 - \frac{c}{t}$ ① intersects hyperbola $xy = c^2$
 at T and V ②

from ② $y = \frac{c^2}{x}$ $\therefore t^2 x - \frac{c^2}{x} = ct^3 - \frac{c}{t}$
 $\frac{(xx)}{(\div t^2)} \quad x^2 - \frac{c^2}{t^2} = xct - \frac{cx}{t^3}$

$$x^2 + \left(\frac{c}{t^3} - dt\right)x - \frac{c^2}{t^2} = 0 \quad \checkmark \quad \text{or } x^2 t^2 + x\left(\frac{c}{t} - dt^3\right) - c^2 = 0$$

$$\cancel{x^2 + \frac{c(1-t^4)}{t^3}x - \frac{c^2}{t^2} = 0} \quad x^2 - dt^3x + \frac{cx}{t^3} - \frac{c^2}{t^2} = 0$$

$$\cancel{\frac{t^3}{t^3}} \quad x(x-dt) + \frac{c}{t^3}(x-dt) = 0$$

$$(x-dt)(x + \frac{c}{t^3}) = 0$$

$$\therefore x = dt \text{ at pt T or } x = -\frac{c}{t^3} \text{ ie at V}$$

$$\text{sub into (2)} \quad -\frac{c}{t^3} \times y = c^2$$

$$y = \frac{c^2 t^3}{-c}$$

$$= -dt^3 \quad \checkmark$$

$$V\left(-\frac{c}{t^3}, -dt^3\right) \text{ as required}$$

(ii) $xy = c^2$
 $x = \frac{c^2}{y}$
 (Alternative)

$$t^2 x - y = dt^3 - \frac{c}{t}$$

$$\frac{c^2 t^2}{y} - y = dt^3 - \frac{c}{t}$$

$$c^2 t^3 - ty^2 = dt^4 y - cy$$

$$ty^2 + (dt^4 - c)y - c^2 t^3 = 0$$

$$y^2 + \frac{c}{t}(t^4 - 1)y - c^2 t^2 = 0$$

$$y(y - \frac{c}{t}) + dt^3(y - \frac{c}{t}) = 0$$

$$(y + dt^3)(y - \frac{c}{t}) = 0$$

$$(iii) P(2ct, 0) \quad Q(0, \frac{2c}{t}) \quad V(-\frac{c}{t^3}, -ct')$$

$$PV^2 = (2ct + \frac{c}{t^3})^2 + (0 + ct^3)^2$$

$$= 4c^2t^2 + \frac{4c^2}{t^2} + \frac{c^2}{t^6} + c^2t^6$$

$$= c^2(t^6 + 4t^2 + \frac{4}{t^2} + \frac{1}{t^6}) \quad \text{or} \quad \frac{c^2}{t^6}(t^{12} + 4t^8 + 4t^4 + 1)$$

$$QV^2 = (0 + \frac{c}{t^3})^2 + (\frac{2c}{t} + ct^3)^2$$

$$= \frac{c^2}{t^6} + \frac{4c^2}{t^2} + 4c^2t^2 + c^2t^6$$

$$= c^2(t^6 + 4t^2 + \frac{4}{t^2} + \frac{1}{t^6})$$

$$\therefore PV^2 = QV^2$$

$PV, QV > 0 \therefore PV = QV$ hence $\triangle PQV$ is isosceles.

OR (easier)

$$\underline{T \text{ is midpoint of } PQ : \quad \left(\frac{2ct+0}{2}, \frac{0+\frac{2c}{t}}{2} \right)}$$

$$= (ct, \frac{c}{t}) \quad (1)$$

$$\therefore TQ = TP$$

TV is common

$$\angle QTV = \angle PTV = 90^\circ$$

$$\therefore \triangle QTV \cong \triangle PTV \text{ (SAS)}$$

(2)

$$\therefore QV = PV \text{ (matching sides in congruent } \triangle s)$$

(14) (a)

$$x = a \cos \theta$$

$$y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\therefore \frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta} \quad (1)$$

$$\therefore \text{gradient of normal} \Rightarrow \frac{a \sin \theta}{b \cos \theta}$$

$$\therefore \text{Eqn normal: } y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta) \quad (2)$$

$$x \cdot \frac{b}{\sin \theta} \Rightarrow \frac{by}{\sin \theta} - b^2 = \frac{ax}{\cos \theta} - a^2$$

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2 \quad (3)$$

(14) (b)

$$(i) \quad \ddot{x} = -kv - g$$

$$v \cdot \frac{dv}{dx} = -(g + kv)$$

$$\begin{aligned} \frac{dx}{dv} &= -\frac{v}{g + kv} = -\frac{1}{k} \left(\frac{g + kv}{g + kv} \right) + \frac{1}{k} \frac{g}{g + kv} \\ &= -\frac{1}{k} + \frac{g}{k^2} \left(\frac{k}{g + kv} \right) \end{aligned}$$

$$\therefore x = -\frac{v}{k} + \frac{g}{k^2} \ln(g + kv) + C, \quad (1)$$

$$\text{now, when } x=0, v=U \quad \therefore C = \frac{U}{k} - \frac{g}{k^2} \ln(g + kU)$$

$$x = \frac{U - v}{k} + \frac{g}{k^2} \ln \left(\frac{g + kv}{g + kU} \right)$$

$$\text{now when } v=0, x=H:$$

$$H = \frac{U}{k} + \frac{g}{k^2} \ln \left(\frac{g}{g + kU} \right) = \frac{U}{k} - \frac{g}{k^2} \ln \left(\frac{g + kU}{g} \right) \quad (2)$$

(14) (b)

(ii) $\ddot{y} = -kw + g$

$$w \cdot \frac{dw}{dy} = g - kw$$

$$\frac{dw}{dy} = \frac{w}{g - kw} = -\frac{1}{k} \frac{g - kw}{g - kw} + \frac{g}{k} \frac{1}{g - kw}$$

$$= -\frac{1}{k} + \frac{g}{k^2} \frac{-k}{g - kw}$$

$$y = -\frac{w}{k} - \frac{g}{k^2} \ln(g - kw) + C_2 \quad (1)$$

when $y=0, w=0 \quad \therefore C_2 = \frac{g}{k^2} \ln g$

$$\therefore y = -\frac{w}{k} + \frac{g}{k^2} \ln\left(\frac{g}{g - kw}\right)$$

now when $y=H, w=W$

$$\therefore H = -\frac{W}{k} + \frac{g}{k^2} \ln\left(\frac{g}{g - kW}\right) \quad (2)$$

(iii) equating H expressions:

$$\frac{U}{k} - \frac{g}{k^2} \ln\left(\frac{g + kU}{g}\right) = -\frac{W}{k} + \frac{g}{k^2} \ln\left(\frac{g}{g - kW}\right)$$

$\times k \downarrow$

$$U + W = \frac{g}{k} \ln\left(\frac{g + kU}{g} \cdot \frac{g}{g - kW}\right)$$

$\div T \text{ or } \times \frac{k}{g} \downarrow$

$$\frac{U}{T} + \frac{W}{T} = \ln\left(\frac{g + kU}{g - kW}\right) \quad (1)$$

$$U_T + W_T = \ln\left(\frac{1 + \frac{U}{\frac{g}{k}}}{1 + \frac{W}{\frac{g}{k}}}\right)$$

$$U_T + W_T = \ln\left(\frac{1 + U_T}{1 - W_T}\right) \quad (2)$$

(14) (b)
(iv)

$$u_T = 50\% \quad w_T = 37\% \quad \text{gives:}$$

$$\text{LHS} = 87\%$$

$$\text{RHS} = \ln \left(\frac{1.5}{0.63} \right) \overset{(1)}{\approx} 86.8\%$$

$$\approx \text{LHS}.$$

(14)

$$(c) \quad x^2 + (y-r)^2 = r^2 \quad y = \left(x^2 - \frac{3}{2}\right)^2$$

$$(i) \text{ if } u = x^2 - \frac{3}{2}$$

$$x^2 = u + \frac{3}{2} \quad \text{and} \quad y = u^2$$

$$\text{so } u + \frac{3}{2} + (u^2 - r)^2 = r^2$$

$$u + \frac{3}{2} + u^4 - 2u^2r + r^2 = r^2$$

$$u^4 - 2u^2r + u + \frac{3}{2} = 0$$

(ii) Now, since the curves are tangent at these points, the eqn in u must have a double root, so the derivative must also give the same values for u .

$$4u^3 - 4ur + 1 = 0$$

(iii) equating (i) gives:

$$x \cdot 4u^2 \cdot \left(\frac{4u^3 + 1}{4u} \right) = \frac{u^4 + u + \frac{3}{2}}{2u^2}$$

$$4u^4 + u = 2u^4 + 2u + 3$$

$$2u^4 - u - 3 = 0$$

$$\text{now } u = -1 \text{ is a solution} \quad \underline{2(1) - (-1) - 3 = 0}$$

Σ

$$(iv) \quad u = -1, \quad r = \frac{4 \times (-1) + 1}{4(-1)} = \frac{3}{4}$$

$$\text{also } r = \frac{1 - 1 + \frac{3}{2}}{2} = \frac{3}{4}$$

$$(\text{Also } x = \pm \frac{1}{\sqrt{2}}, y = 1)$$

(Another ^{real} solution exists, $u \approx 1.204$, $r \approx 1.657$, $(x, y) \approx (\pm 1.644, 1.450)$ but it is outside the given domain.